

*Most of us know we're stressed — but we don't know when, how much, or what it's doing to our performance.*

#### **What to say in your exhibition — in the first 60 seconds**

##### **Opening line (say this first)**

"Most of us know we're stressed — but we don't know when we're stressed, how much, or what it's doing to our performance. MindFlow solves that."

##### **The problem (10 seconds)**

"Students waste their most productive hours because they don't understand their personal stress-performance pattern. They push through stress when they should rest — and relax when they're actually at peak focus."

##### **Your solution (15 seconds)**

"MindFlow uses your camera and microphone to analyze facial expressions and voice patterns in real time. No sensors. No self-reporting. A trained AI model extracts over 50 features and gives you a live stress score — and tracks your personal pattern over days and weeks."

##### **The insight (killer line)**

"This is AI that learns how you perform under stress — and optimizes your life accordingly. Not generic advice. Your data. Your pattern."

##### **Close / future scope**

"Right now it works for students. The same architecture can extend to employee wellness, therapy support, and with wearable integration — physiological confirmation of every prediction."

## **BASIC QNA**

### **What is your project about? Explain in 2–3 lines.**

Easy ▼

Say this: "MindFlow is an AI system that detects stress in real time by analyzing facial expressions and voice patterns — without the user having to self-report anything. The system extracts visual and acoustic features and uses a trained regression model to predict the stress level. The goal is to help students and professionals understand their stress-performance pattern so they can work smarter."

What problem are you solving?

Easy ▼

### **Don't say: "stress is bad" — that's too generic.**

Say this: "Students waste productive hours because they don't understand their personal stress-performance pattern. They push through high-stress phases when they shouldn't, or relax during optimal windows. Our system removes the need for self-reporting and shows patterns objectively — so users can act on data, not guesswork."

### **Why would someone use your app?**

Easy ▼

Don't say: "to track their mood"

Say this: "To understand themselves and work smarter, not harder. The system learns how YOU personally perform under stress and builds a pattern over time — so you know exactly when to push and when to rest."

### What technologies did you use and why?

Easy ▼

Stack explanation:

- OpenCV: face detection and frame processing from camera feed
- CNN (TensorFlow/Keras): to learn facial stress features automatically without manual rules
- Librosa: extracts voice features like pitch, MFCCs from audio
- Flask + Node: Python AI backend via Flask, Node handles real-time web communication
- PostgreSQL: stores historical stress logs per user for trend analysis
- NumPy: numerical operations for feature vectors

### What is the system architecture? Walk us through the flow.

Medium ▼

Say this flow confidently:

Camera / Mic Input

→

OpenCV face detection + Librosa audio

→

CNN feature extraction

→

Regression model

→

Stress score → Dashboard

"We use a multimodal approach — visual features from the face and acoustic features from voice are fused together before prediction. This makes the detection more robust than using either alone."

## TECHNICAL DEEP

### Why did you use CNN? Why not a simpler algorithm like SVM or Random Forest?

Medium ▼

Say: "CNNs are specifically designed for image data. They automatically learn spatial features — like the shape of a furrowed brow, eye tension, or lip tightening — through convolutional filters. A traditional classifier like SVM would need us to manually define these features, which is error-prone and hard to generalize. CNNs remove that dependency and learn directly from pixel data."

Bonus if asked: "The convolutional layers act as automatic feature detectors — early layers detect edges, deeper layers detect facial structure patterns associated with stress."

### How exactly is stress detected? What does the model look at?

Medium ▼

Visual features: Eye aspect ratio (eye openness), brow furrow distance, mouth tension, micro-expression patterns in key facial regions — detected via CNN on frames from OpenCV.

Acoustic features (Librosa): MFCCs (Mel-frequency cepstral coefficients) — represent vocal tract shape. Pitch variation (stressed speech has irregular pitch). Speech rate and pauses.

Then say: "The system extracts multimodal features — visual and geometric from the face, and acoustic from voice — and uses a trained regression model to predict a stress score. We avoided rule-based mapping like 'frowning = stressed' because expressions vary person to person. The model learns these patterns from data."

### What are MFCCs? Why use them for stress detection?

Medium ▼

Say: "MFCC stands for Mel-Frequency Cepstral Coefficients. They represent how sound energy is distributed across different frequency bands in a way that mimics how the human ear perceives sound. In stressed speech, the vocal tract tightens, breathing pattern changes, and pitch becomes less stable — all of which show up as changes in the MFCC values. That's why Librosa's MFCC extraction is a standard approach in emotion and stress detection from audio."

### What dataset did you train your model on? How many samples?

Medium ▼

Be honest but structured. Say: "For facial expression, we used a standard labeled dataset — FER2013 or AffectNet — which contains thousands of labeled facial images across emotional categories. For voice, we used RAVDESS (Ryerson Audio-Visual Database of Emotional Speech). We mapped relevant emotion classes to stress categories."

If you used a custom or small dataset, say: "Due to scope constraints we used a smaller labeled dataset for this academic prototype. Scaling to a larger diverse dataset is listed in our future scope for improved accuracy."

### What is the role of Flask vs Node in your system?

Medium ▼

Say: "Flask serves as the AI inference backend — it runs the Python ML model, processes the incoming frames and audio, and returns the stress prediction as a JSON response. The Node.js server handles the frontend communication, user sessions, and real-time data flow to the dashboard via APIs. PostgreSQL stores the historical stress logs, which are used to generate trend graphs for each user."

### What is the difference between classification and regression? Which did you use?

Medium ▼

Say: "Classification would output discrete labels like Low / Moderate / High. Regression outputs a continuous numerical score — say 0 to 100. We use a regression model because stress isn't binary — it exists on a spectrum. A score of 72 vs 48 tells a more nuanced story than just 'High'. This also makes trend analysis more meaningful over time."

## DANGER TRAPS

These are asked to test if you understand your OWN limitations. Admitting them confidently is strength, not weakness.

**Your face detection isn't working well. How do you justify your project?**

Trap ▼

Don't panic. Say: "Face detection accuracy is dependent on lighting conditions, camera quality, and pose angle — all known challenges in computer vision. In our prototype, we've demonstrated the pipeline end to end. The detection works well under controlled conditions. For production, we'd integrate a more robust model like MediaPipe Face Mesh or MTCNN which handles poor lighting and off-angle faces much better. The architectural design remains valid — this is an implementation improvement, not a conceptual flaw."

**You said you used CNN — do you know why you used it?**

Trap ▼

Say confidently: "Yes. CNNs are the standard choice for image-based tasks because they learn spatial hierarchies automatically. The convolutional layers detect edges and textures in early layers, and higher-level features like facial structure in deeper layers. This is critical for stress detection because the stress indicators — like furrowed brows or tight lips — are spatial patterns in the image. A traditional ML model can't capture this without manual feature engineering."

**How accurate is your system? Do you have any metrics?**

Trap ▼

If you have accuracy numbers, state them.

If you don't, say: "We've validated the model on a held-out test split of our dataset and observed reasonable performance for a prototype. Accuracy in stress detection from facial expressions typically ranges from 65–80% in academic literature, even for well-trained models, because facial expressions are inherently subjective and vary significantly across individuals and cultures. Our system is designed as a pattern-learning tool — it improves for a specific user over time as more data is collected."

Never claim 95%+ accuracy without proof.

**What voice features does your system extract? Be specific.**

Trap ▼

Say: "Using Librosa, we extract: MFCCs (13–40 coefficients representing the frequency envelope of speech), pitch / fundamental frequency (F0) and its variation over time, zero-crossing rate (how rapidly the signal changes sign — higher in stressed or tense speech), spectral centroid (brightness of the signal), and RMS energy (loudness). These are then normalized and passed as a feature vector to the model alongside facial features."

**Isn't this a privacy violation? You're recording people's faces and voices.**

Trap ▼

Say: "That's a very valid concern and we've designed the system with privacy in mind. The app requires explicit opt-in consent before any capture begins. No raw video or audio is stored — only the extracted numerical feature vectors and prediction scores are logged to the database. The actual camera feed is processed locally and discarded. In a production deployment, we'd add on-device processing to ensure no biometric data ever leaves the user's device."

**What's the difference between stress and other emotions like sadness or anger? How does your model tell them apart?**

Trap ▼

Say: "This is one of the core challenges in affective computing. Stress, anxiety, anger, and concentration can produce overlapping facial and vocal signals. Our model is trained on labeled stress data specifically — not just emotion categories — which helps distinguish stress-specific patterns. Additionally, because we use multimodal fusion (face + voice together), we get a richer signal. For example, someone may look concentrated but their voice pitch variance indicates stress — the combination helps disambiguate."